

Answer Key

ΘEIII Mathematics — Paper Diagnostic · 30 items · every answer computed and machine-verified

Not for students. Option order on the question sheet is shuffled per item; the letters below are the letters as printed. If the question sheet is ever regenerated, regenerate this key with it.

ALGEBRA

NO.	ANS	TIER	SKILL	WORKING
1	C	1	integer-operations	Multiplication first: $5 \times (-2) = -10$, then $-3 + (-10) = -13$.
2	B	1	fraction-arithmetic	Common denominator 12: $\frac{8}{12} + \frac{3}{12} = \frac{11}{12}$.
3	A	1	linear-equation-one-variable	$2x = 16$, so $x = 8$.
4	D	1	exponent-rules	Add the exponents on top, subtract the one below: $x^{5+3-4} = x^4$.
5	C	2	factoring-quadratic-trinomial	Two numbers multiplying to 12 and adding to -7 : -3 and -4 .
6	B	2	systems-two-linear	Adding the equations gives $3x = 12$, so $x = 4$ and then $y = 3$.
7	C	2	composite-functions	$g(3) = 9$, then $f(9) = 2 \cdot 9 + 1 = 19$.
8	D	3	logarithmic-equations	$\log_2(x(x-2)) = 3$ gives $x^2 - 2x - 8 = 0$, so $x = 4$ or $x = -2$. Only $x = 4$ keeps both logarithms defined.
9	A	3	vieta-formulas	For $ax^2 + bx + c = 0$ the sum of the roots is $-\frac{b}{a} = \frac{6}{2} = 3$.
10	C	3	quadratic-inequalities	$(x-2)(x-3) < 0$, so x lies strictly between the roots: $2 < x < 3$.

GEOMETRY & TRIGONOMETRY

NO.	ANS	TIER	SKILL	WORKING
11	A	1	angles-and-parallel-lines	Co-interior angles are supplementary: $180^\circ - 65^\circ = 115^\circ$.
12	D	1	pythagoras	$\sqrt{5^2 + 12^2} = \sqrt{169} = 13$.
13	A	1	circle-area-and-arc	$A = \pi r^2 = 36\pi$.
14	B	2	triangle-similarity	Areas scale by the square of the ratio: $18 \times \left(\frac{5}{3}\right)^2 = 50$.
15	A	2	exact-trig-values	$\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} = \frac{3}{4}$.
16	D	2	prism-volume-surface	$2(3 \cdot 4 + 4 \cdot 5 + 3 \cdot 5) = 2(12 + 20 + 15) = 94$.
17	C	3	cone	Height $= \sqrt{5^2 - 3^2} = 4$, so $V = \frac{1}{3}\pi r^2 h = 12\pi$.
18	B	3	dot-product	$3(2) + 4(-1) = 6 - 4 = 2$.

ANALYSIS

NO.	ANS	TIER	SKILL	WORKING
19	C	1	derivative-power-rule	$\frac{d}{dx}(3x^4) = 12x^3$ and $\frac{d}{dx}(-2x) = -2$.
20	D	1	antiderivative-power	Raise the exponent and divide: $\frac{6x^3}{3} = 2x^3$, plus C .
21	C	2	tangent-line	At $x = 2$: $y = -2$ and $y' = 2x - 3 = 1$. So $y + 2 = 1(x - 2)$, i.e. $y = x - 4$.
22	B	2	definite-integral	$[x^3]_0^2 = 8 - 0 = 8$.
23	C	3	stationary-points-and-extrema	$f'(x) = 3x^2 - 3 = 0$ at $x = \pm 1$. The maximum is at $x = -1$, where $f(-1) = 2$.

PROBABILITY & STATISTICS

NO.	ANS	TIER	SKILL	WORKING
24	B	1	mean-median-mode	Ordered: 3, 7, 7, 9, 14. The middle value is 7.
25	A	1	classical-probability	$P = \frac{\text{red}}{\text{total}} = \frac{3}{8}$.
26	D	2	quartiles-and-iqr	$Q_1 = 4$ and $Q_3 = 10$, so $\text{IQR} = 10 - 4 = 6$.
27	A	2	addition-rule	$P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$.
28	B	3	expected-value	$E(X) = 0(0.2) + 1(0.5) + 2(0.3) = 1.1$.

COMBINATORICS

NO.	ANS	TIER	SKILL	WORKING
29	A	1	counting-principle	$4 \times 3 = 12$.
30	D	2	combinations	$\binom{8}{3} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56$.

What each wrong answer means

Every distractor was built from one specific mistake. A class that picks the same wrong option is a class with one teachable gap, not a class that guessed — so this table is worth reading before you re-teach anything.

ALGEBRA

NO.	OPTION	THE MISTAKE IT COMES FROM
1	A	lost the sign of the product
1	B	sign slip on the final answer
1	D	added before multiplying
2	A	converted the first fraction wrongly ($\frac{4}{12}$)
2	C	added numerators and denominators across
2	D	converted the second fraction to twelfths wrongly ($\frac{1}{12}$)
3	B	moved both constants the same way ($2x = 2$)
3	C	sign slip moving the constant
3	D	did not divide by the coefficient
4	A	did not subtract the denominator's exponent
4	B	added the denominator's exponent instead of subtracting
4	C	multiplied the exponents instead of adding
5	A	product correct but sum is -8
5	B	product correct but sum is -13
5	D	right numbers, both signs wrong
6	A	subtracted the equations instead of adding
6	C	wrote the solution in the wrong order
6	D	sign slip solving for y
7	A	stopped after the inner function
7	B	computed $g(f(3))$ — applied the functions in the wrong order
7	D	dropped the coefficient 2
8	A	did not reject the root outside the domain
8	B	added the arguments instead of multiplying them
8	C	kept the rejected root and discarded the valid one
9	B	gave the product c/a instead of the sum
9	C	did not divide by a
9	D	kept the minus sign in $-b/a$ after b was already negative
10	A	included the endpoints although the inequality is strict
10	B	took the outside of the roots — the wrong side of the parabola
10	D	sign error when reading the roots off the factors

GEOMETRY & TRIGONOMETRY

NO.	OPTION	THE MISTAKE IT COMES FROM
11	B	used complementary instead of supplementary
11	C	used the alternate-angle rule instead of co-interior
11	D	doubled the angle
12	A	added the legs
12	B	treated 12 as the hypotenuse
12	C	subtracted the legs
13	B	dropped the π
13	C	used πr instead of πr^2
13	D	used the circumference formula
14	A	used $\frac{5}{2}$ as the scale factor
14	C	scaled by the ratio instead of its square
14	D	multiplied by 5
15	B	gave one factor instead of the product
15	C	used $\sin 30^\circ \cos 60^\circ$
15	D	used $\sin 30^\circ$ for both
16	A	found the volume instead
16	B	counted each face once
16	C	doubled the volume
17	A	used $\pi r^2 h$ without the $\frac{1}{3}$
17	B	both errors together
17	D	used the slant height as the perpendicular height
18	A	added the components of $\vec{d}$
18	C	ignored the negative component
18	D	sign slip on the whole sum

ANALYSIS

NO.	OPTION	THE MISTAKE IT COMES FROM
19	A	did not bring the exponent down as a factor
19	B	left the linear term undifferentiated
19	D	did not reduce the exponent
20	A	differentiated instead of integrating
20	B	did not divide at all
20	C	divided by the old exponent

21	A	used the y -value as the gradient
21	B	used the x -value as the intercept
21	D	sign slip substituting the point
22	A	substituted into the integrand rather than the antiderivative
22	C	used $\frac{1}{2}bh$ on the curve
22	D	divided by the exponent twice
23	A	gave the x -coordinate of the maximum
23	B	gave the local minimum value
23	D	gave the x -coordinate of the minimum

PROBABILITY & STATISTICS

NO.	OPTION	THE MISTAKE IT COMES FROM
24	A	took the fourth value
24	C	computed the mean
24	D	took the middle of the unordered list
25	B	used the number of colours as the total
25	C	gave $P(\text{blue})$
25	D	used red-to-blue instead of red-to-total
26	A	gave the range
26	B	gave the median
26	C	used the wrong quartile positions
27	B	treated the events as mutually exclusive
27	C	gave the intersection
27	D	added the intersection instead of subtracting
28	A	gave the largest probability
28	C	averaged the values, ignoring the probabilities
28	D	averaged only the non-zero values

COMBINATORICS

NO.	OPTION	THE MISTAKE IT COMES FROM
29	B	added instead of multiplying
29	C	doubled the product
29	D	computed 4^3
30	A	computed 8×3
30	B	forgot to divide by $3!$ fully

30	C	counted arrangements (8P_3) — order should not matter
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